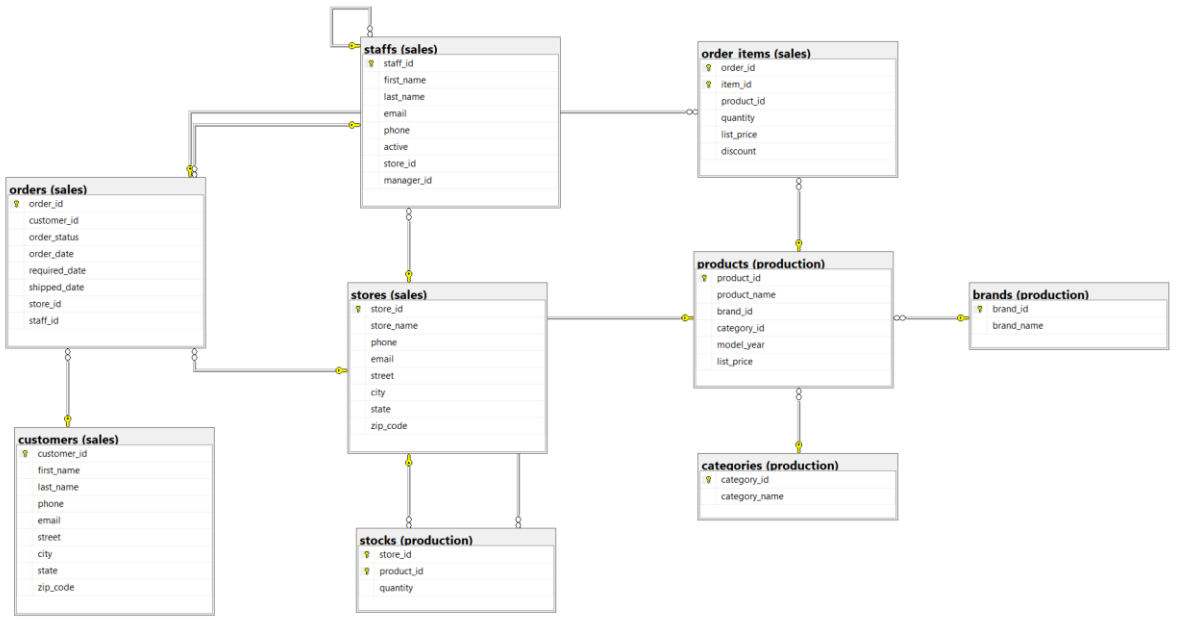
Tarallo Luca

22/04/2026

Diagramma –



Queries :

1. Selezionare i prodotti che hanno una certa marca

```
Select
        p.product_id,
        p.product_name,
        p.list_price,
        p.model_year,
        b.brand_name
from production.products p
join production.brands b on p.brand_id = b.brand_id
where b.brand_name = 'Trek';
```

Risultati | Messaggi

	product_id	product_name	list_price	model_year	brand_name
1	1	Trek 820 - 2016	379.99	2016	Trek
2	4	Trek Fuel EX 8 29 - 2016	2899.99	2016	Trek
3	7	Trek Slash 8 27.5 - 2016	3999.99	2016	Trek
4	8	Trek Remedy 29 Carbon Frameset - 2016	1799.99	2016	Trek
5	9	Trek Conduit+ - 2016	2999.99	2016	Trek
6	29	Trek X-Caliber 8 - 2017	999.99	2017	Trek
7	32	Trek Farley Alloy Frameset - 2017	469.99	2017	Trek
8	34	Trek Session DH 27.5 Carbon Frameset - 2017	469.99	2017	Trek
9	39	Trek Stache 5 - 2017	1499.99	2017	Trek

Ese... | CSP-2527-14 (17.0 RTM) | EDU-ITS\luca.tarallo (55) | BikeStores | 00:00:00 | Riga: 1, Colonna: 1 | Righe 135

2. Selezionare i prodotti che hanno in stocks almeno 20 pezzi

```
Select
    p.product_id,
    p.product_name,
    b.brand_name,
    s.store_id,
    s.store_name,
    st.quantity
from production.products p
join production.brands b  on p.brand_id  = b.brand_id
join production.stocks st on p.product_id = st.product_id
join sales.stores s       on st.store_id  = s.store_id
where st.quantity >= 20
order by st.quantity desc;
```

Risultati | Messaggi

	product_id	product_name	brand_name	store_id	store_name	quantity
1	11	Surly Straggler 650b - 2016	Surly	3	Rowlett Bikes	30
2	12	Electra Townie Original 21D - 2016	Electra	3	Rowlett Bikes	30
3	30	Surly Ice Cream Truck Frameset - 2017	Surly	1	Santa Cruz Bikes	30
4	37	Haro Flightline One ST - 2017	Haro	3	Rowlett Bikes	30
5	61	Trek Powerfly 8 FS Plus - 2017	Trek	1	Santa Cruz Bikes	30
6	64	Electra Townie Original 7D - 2017	Electra	1	Santa Cruz Bikes	30
7	64	Electra Townie Original 7D - 2017	Electra	2	Baldwin Bikes	30
8	68	Sun Bicycles Cruz 3 - 2017	Sun Bicycles	1	Santa Cruz Bikes	30
9	71	Sun Bicycles Atlas X-Type - 2017	Sun Bicycles	2	Baldwin Bikes	30

Ese... | CSP-2527-14 (17.0 RTM) | EDU-ITS\luca.tarallo (55) | BikeStores | 00:00:00 | Riga: 1, Colonna: 1 | Righe 311

3. Selezionare gli ordini gestiti da un certo componente dello staff (cognome)

```
Select
    o.order_id,
    o.order_date,
    o.required_date,
```

```
        o.shipped_date,
        o.order_status,
        st.last_name
from sales.orders o
join sales.staffs st on o.staff_id = st.staff_id
where st.last_name = 'Copeland'
order by o.order_date desc;
```

Risultati | Messaggi

	order_id	order_date	required_date	shipped_date	order_status	last_name
1	1608	2018-07-12	2018-07-12	NULL	3	Copeland
2	1607	2018-07-11	2018-07-11	NULL	3	Copeland
3	1591	2018-04-27	2018-04-27	NULL	2	Copeland
4	1586	2018-04-26	2018-04-26	NULL	1	Copeland
5	1587	2018-04-26	2018-04-26	NULL	2	Copeland
6	1581	2018-04-25	2018-04-25	NULL	2	Copeland
7	1579	2018-04-24	2018-04-24	NULL	2	Copeland
8	1580	2018-04-24	2018-04-24	NULL	2	Copeland
9	1573	2018-04-23	2018-04-23	NULL	2	Copeland

Ese... | CSP-2527-14 (17.0 RTM) | EDU-ITS\luca.tarallo (55) | BikeStores | 00:00:00 | Riga: 1, Colonna: 1 | Righe 164

4. Selezionare il totale delle vendite del primo semestre 2016

```
Select
sum(oi.quantity * oi.list_price * (1 - oi.discount)) as totale_vendite,
count(distinct o.order_id) as numero_ordini
from sales.orders o
join sales.order_items oi on o.order_id = oi.order_id
where o.order_date >= '2016-01-01'
and o.order_date <  '2016-07-01';
```

Risultati | Messaggi

	totale_vendite	numero_ordini
1	1134835.2602	293

Esecu... | CSP-2527-14 (17.0 RTM) | EDU-ITS\luca.tarallo (55) | BikeStores | 00:00:00 | Riga: 1, Colonna: 1 | Righe 1

5. Selezionare i clienti che hanno acquistato prodotti di una certa categoria (nome)

```
Select distinct
    c.customer_id,
    c.first_name,
    c.last_name,
    c.email,
    cat.category_name
from sales.customers c
join sales.orders     o   on c.customer_id  = o.customer_id
join sales.order_items oi on o.order_id     = oi.order_id
join production.products p  on oi.product_id  = p.product_id
join production.categories cat on p.category_id = cat.category_id
where cat.category_name = 'Mountain Bikes'
order by c.last_name, c.first_name;
```

Risultati | Messaggi

	customer_id	first_name	last_name	email	category_name
1	1445	Ester	Acevedo	ester.acevedo@gmail.com	Mountain Bikes
2	61	Elinore	Aguilar	elinore.aguilar@msn.com	Mountain Bikes
3	39	Janetta	Aguirre	janetta.aguirre@aol.com	Mountain Bikes
4	1154	Barry	Albert	barry.albert@gmail.com	Mountain Bikes
5	1083	Cher	Alston	cher.alston@msn.com	Mountain Bikes
6	1402	Sandee	Alvarado	sandee.alvarado@aol.com	Mountain Bikes
7	1134	Sherril	Alvarado	sherril.alvarado@hotmail.com	Mountain Bikes
8	1120	Karole	Alvarez	karole.alvarez@gmail.com	Mountain Bikes
9	327	Sharie	Alvarez	sharie.alvarez@msn.com	Mountain Bikes

Ese... | CSP-2527-14 (17.0 RTM) | EDU-ITS\luca.tarallo (55) | BikeStores | 00:00:00 | Riga: 1, Colonna: 1 | Righe 810

Views :

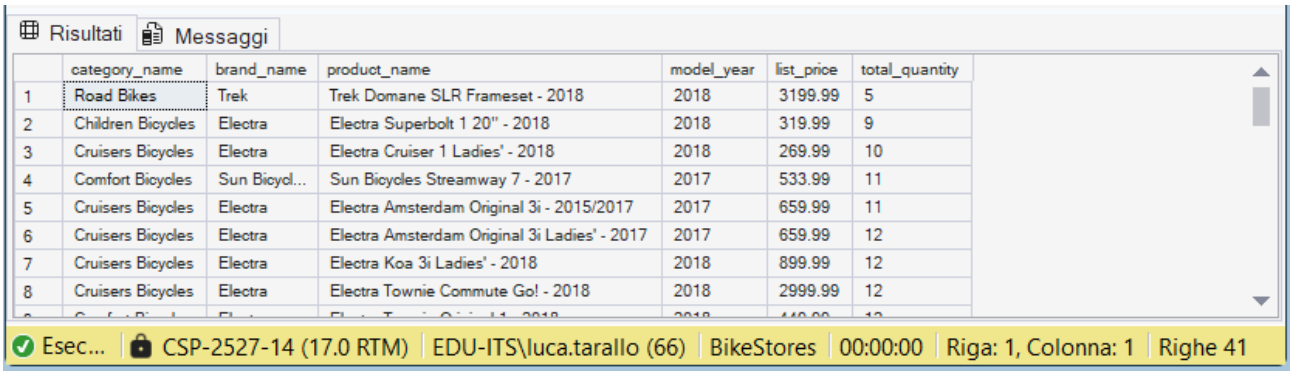
1. Creare la vista vw_ProductionByQuantity con category_name, brand_name, product_name, model_year, list_price, quantity dello schema production. Interrogare la view richiedendo solo i dati con quantity strettamente inferiori a 25 unità

    - Creazione della View :

```
Create view production.vw_ProductionByQuantity as
Select
    cat.category_name,
    b.brand_name,
    p.product_name,
    p.model_year,
    p.list_price,
    sum(st.quantity) as total_quantity
from production.products p
join production.categories cat on p.category_id = cat.category_id
join production.brands b on p.brand_id = b.brand_id
join production.stocks st on p.product_id = st.product_id
group by
    cat.category_name,
    b.brand_name,
    p.product_name,
    p.model_year,
    p.list_price;
```

    - Interrogazione della View :

```
Select *
from production.vw_ProductionByQuantity
where total_quantity < 25
order by total_quantity asc;
```



Risultati | Messaggi

	category_name	brand_name	product_name	model_year	list_price	total_quantity
1	Road Bikes	Trek	Trek Domane SLR Frameset - 2018	2018	3199.99	5
2	Children Bicycles	Electra	Electra Superbolt 1 20" - 2018	2018	319.99	9
3	Cruisers Bicycles	Electra	Electra Cruiser 1 Ladies' - 2018	2018	269.99	10
4	Comfort Bicycles	Sun Bicycl...	Sun Bicycles Streamway 7 - 2017	2017	533.99	11
5	Cruisers Bicycles	Electra	Electra Amsterdam Original 3i - 2015/2017	2017	659.99	11
6	Cruisers Bicycles	Electra	Electra Amsterdam Original 3i Ladies' - 2017	2017	659.99	12
7	Cruisers Bicycles	Electra	Electra Koa 3i Ladies' - 2018	2018	899.99	12
8	Cruisers Bicycles	Electra	Electra Townie Commute Go! - 2018	2018	2999.99	12

Esec... | CSP-2527-14 (17.0 RTM) | EDU-ITS\luca.tarallo (66) | BikeStores | 00:00:00 | Riga: 1, Colonna: 1 | Righe 41

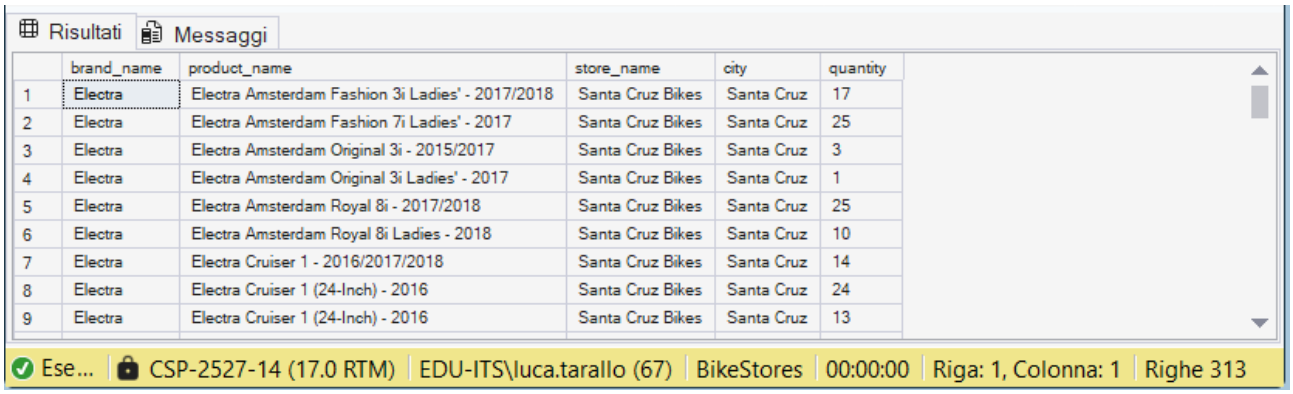
2. Creare la vista vw_StoresByQuantity con brand_name, product_name, store_name, city, quantity. Interrogare la view per visualizzare i dati solo di una certa città

   - Creazione della View :

```
Create view sales.vw_StoresByQuantity as
Select
   b.brand_name,
   p.product_name,
   s.store_name,
   s.city,
   st.quantity
from production.products p
join production.brands   b  on p.brand_id   = b.brand_id
join production.stocks   st on p.product_id  = st.product_id
join sales.stores        s  on st.store_id   = s.store_id;
```

   - Interrogazione della View :

```
Select *
from sales.vw_StoresByQuantity
where city = 'Santa Cruz'
order by brand_name, product_name;
```



Risultati | Messaggi

	brand_name	product_name	store_name	city	quantity
1	Electra	Electra Amsterdam Fashion 3i Ladies' - 2017/2018	Santa Cruz Bikes	Santa Cruz	17
2	Electra	Electra Amsterdam Fashion 7i Ladies' - 2017	Santa Cruz Bikes	Santa Cruz	25
3	Electra	Electra Amsterdam Original 3i - 2015/2017	Santa Cruz Bikes	Santa Cruz	3
4	Electra	Electra Amsterdam Original 3i Ladies' - 2017	Santa Cruz Bikes	Santa Cruz	1
5	Electra	Electra Amsterdam Royal 8i - 2017/2018	Santa Cruz Bikes	Santa Cruz	25
6	Electra	Electra Amsterdam Royal 8i Ladies - 2018	Santa Cruz Bikes	Santa Cruz	10
7	Electra	Electra Cruiser 1 - 2016/2017/2018	Santa Cruz Bikes	Santa Cruz	14
8	Electra	Electra Cruiser 1 (24-Inch) - 2016	Santa Cruz Bikes	Santa Cruz	24
9	Electra	Electra Cruiser 1 (24-Inch) - 2016	Santa Cruz Bikes	Santa Cruz	13

Ese... | CSP-2527-14 (17.0 RTM) | EDU-ITS\luca.tarallo (67) | BikeStores | 00:00:00 | Riga: 1, Colonna: 1 | Righe 313

3. Creare la vista vw_ProductsByCustomers con product_name, brand_name, category_name, quantity (quantità di acquisto), list_price (prezzo finale di acquisto), discount (sconto di acquisto), order_date, first_name, last_name, city. Interrogare la view vw_ProductsByCustomers per visualizzare i prodotti ordinati in una certa data

   - Creazione della View :

```
Create view sales.vw_ProductsByCustomers as
Select
```

```
    p.product_name,
    b.brand_name,
    cat.category_name,
    oi.quantity,
    oi.list_price,
    oi.discount,
    o.order_date,
    c.first_name,
    c.last_name,
    c.city
from sales.order_items    oi
join sales.orders         o   on oi.order_id    = o.order_id
join sales.customers      c   on o.customer_id  = c.customer_id
join production.products  p   on oi.product_id  = p.product_id
join production.brands    b   on p.brand_id     = b.brand_id
join production.categories cat on p.category_id = cat.category_id;
```

- Interrogazione della View :

```
Select *
from sales.vw_ProductsByCustomers
where order_date = '2016-01-01'
order by last_name, first_name;
```

Risultati | Messaggi

	product_name	brand_name	category_name	quantity	list_price	discount	order_date	first_name	last_name	cit
1	Electra Townie Original 7D EQ - Women's - 2016	Electra	Cruisers Bicycles	1	599.99	0.07	2016-01-...	Jaqueline	Cummin...	H
2	Electra Townie Original 7D EQ - 2016	Electra	Cruisers Bicycles	2	599.99	0.05	2016-01-...	Jaqueline	Cummin...	H
3	Electra Townie Original 7D EQ - Women's - 2016	Electra	Cruisers Bicycles	1	599.99	0.20	2016-01-...	Johnath...	Velazqu...	P
4	Trek Remedy 29 Carbon Frameset - 2016	Trek	Mountain Bikes	2	1799.99	0.07	2016-01-...	Johnath...	Velazqu...	P
5	Surly Straggler - 2016	Surly	Cyclocross Bicycles	2	1549.00	0.05	2016-01-...	Johnath...	Velazqu...	P
6	Electra Townie Original 7D EQ - 2016	Electra	Cruisers Bicycles	2	599.99	0.05	2016-01-...	Johnath...	Velazqu...	P
7	Trek Fuel EX 8 29 - 2016	Trek	Mountain Bikes	1	2899.99	0.20	2016-01-...	Johnath...	Velazqu...	P

Esecu... | CSP-2527-14 (17.0 RTM) | EDU-ITS\luca.tarallo (61) | BikeStores | 00:00:00 | Riga: 1, Colonna: 1 | Righe 7

Stored Procedures :

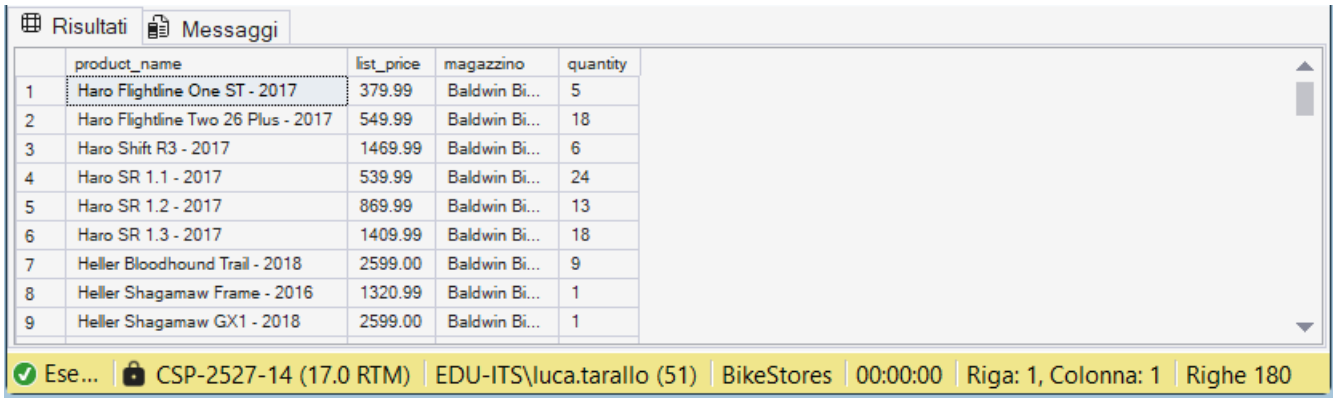
1. Selezionare il prodotto, il prezzo, il magazzino e le quantità dei prodotti di una certa categoria

- Stored Procedure :

```
Create procedure production.usp_GetProductsByCategory
    @category_name varchar(100)
as
begin
    Select
        p.product_name,
        p.list_price,
        s.store_name    as magazzino,
        st.quantity
    from production.products   p
    join production.categories cat on p.category_id =
cat.category_id
    join production.stocks     st  on p.product_id  =
st.product_id
    join sales.stores          s   on st.store_id   = s.store_id
    where cat.category_name = @category_name
    order by s.store_name, p.product_name;
end;
```

- Esecuzione Stored Procedure :

```
exec production.usp_GetProductsByCategory @category_name = 'Mountain Bikes';
```



Risultati | Messaggi

	product_name	list_price	magazzino	quantity
1	Haro Flightline One ST - 2017	379.99	Baldwin Bi...	5
2	Haro Flightline Two 26 Plus - 2017	549.99	Baldwin Bi...	18
3	Haro Shift R3 - 2017	1469.99	Baldwin Bi...	6
4	Haro SR 1.1 - 2017	539.99	Baldwin Bi...	24
5	Haro SR 1.2 - 2017	869.99	Baldwin Bi...	13
6	Haro SR 1.3 - 2017	1409.99	Baldwin Bi...	18
7	Heller Bloodhound Trail - 2018	2599.00	Baldwin Bi...	9
8	Heller Shagamaw Frame - 2016	1320.99	Baldwin Bi...	1
9	Heller Shagamaw GX1 - 2018	2599.00	Baldwin Bi...	1

Ese... | CSP-2527-14 (17.0 RTM) | EDU-ITS\luca.tarallo (51) | BikeStores | 00:00:00 | Riga: 1, Colonna: 1 | Righe 180

2. Selezionare il nome, il prezzo, la categoria, la quantità e la marca dei prodotti che hanno in magazzino almeno n pezzi

- Stored Procedure :

```
Create procedure production.usp_GetProductsByMinStock
    @min_quantity int
as
begin
    Select
        p.product_name,
        p.list_price,
        cat.category_name,
        sum(st.quantity) as quantity,
        b.brand_name
    from production.products   p
    join production.categories cat on p.category_id =
cat.category_id
    join production.brands     b   on p.brand_id    = b.brand_id
    join production.stocks     st  on p.product_id  =
st.product_id
    group by
        p.product_name,
        p.list_price,
        cat.category_name,
        b.brand_name
    having sum(st.quantity) >= @min_quantity
    order by quantity desc;
end;
```

- Esecuzione Stored Procedure :

```
exec production.usp_GetProductsByMinStock @min_quantity = 20;
```

Risultati | Messaggi

	product_name	list_price	category_name	quantity	brand_name
1	Trek XM700+ Lowstep - 2018	3499.99	Electric Bikes	86	Trek
2	Electra Townie Original 7D - 2017	489.99	Cruisers Bicycles	82	Electra
3	Sun Bicycles Cruz 7 - Women's - 2017	416.99	Comfort Bicycles	79	Sun Bicycles
4	Trek Verve+ - 2018	2299.99	Electric Bikes	79	Trek
5	Trek Powerfly 8 FS Plus - 2017	4999.99	Electric Bikes	78	Trek
6	Trek Domane AL 2 Women's - 2018	749.99	Road Bikes	77	Trek
7	Trek Domane SL 5 Disc - 2018	2499.99	Road Bikes	77	Trek
8	Trek Fuel EX 8 29 XT - 2018	3199.99	Mountain Bikes	75	Trek
9	Electra Townie Commute Go! Ladies' - 2018	2999.99	Cruisers Bicycles	75	Electra

Ese... | CSP-2527-14 (17.0 RTM) | EDU-ITS\luca.tarallo (51) | BikeStores | 00:00:00 | Riga: 1, Colonna: 1 | Righe 288

3. Selezionare il nome, il cognome, l'email e il telefono degli impiegati che lavorano in un certo negozio individuato tramite il nome

   - Stored Procedure :

```
Create procedure sales.usp_GetStaffByStoreName
    @store_name varchar(100)
as
begin
    Select
        st.first_name,
        st.last_name,
        st.email,
        st.phone
    from sales.staffs st
    join sales.stores s on st.store_id = s.store_id
    where s.store_name = @store_name
    order by st.last_name, st.first_name;
end;
```

   - Esecuzione Stored Procedure :

```
exec sales.usp_GetStaffByStoreName @store_name = 'Santa Cruz Bikes';
```

Risultati | Messaggi

	first_name	last_name	email	phone
1	Mireya	Copeland	mireya.copeland@bikes.shop	(831) 555-5555
2	Fabiola	Jackson	fabiola.jackson@bikes.shop	(831) 555-5554
3	Genna	Serrano	genna.serrano@bikes.shop	(831) 555-5556
4	Virgie	Wiggins	virgie.wiggins@bikes.shop	(831) 555-5557

Esecu... | CSP-2527-14 (17.0 RTM) | EDU-ITS\luca.tarallo (51) | BikeStores | 00:00:00 | Riga: 1, Colonna: 1 | Righe 4

4. Selezionare l'id, la data dell'ordine, la data di richiesta e di spedizione, lo stato degli ordini gestiti da un certo impiegato individuato tramite il cognome e il nome

   - Stored Procedure :

```
Create procedure sales.usp_GetOrdersByStaff
    @last_name  varchar(100),
```

```
    @first_name varchar(100)
as
begin
    Select
        o.order_id,
        o.order_date,
        o.required_date,
        o.shipped_date,
        o.order_status
    from sales.orders  o
    join sales.staffs  st on o.staff_id = st.staff_id
    where st.last_name  = @last_name
      and st.first_name = @first_name
    order by o.order_date desc;
end;
```

- Esecuzione Stored Procedure :

```
exec sales.usp_GetOrdersByStaff @last_name  = 'Serrano', @first_name = 'Genna';
```

Risultati | Messaggi

	order_id	order_date	required_date	shipped_date	order_status
1	1612	2018-10-21	2018-10-21	NULL	3
2	1598	2018-04-29	2018-04-29	NULL	2
3	1589	2018-04-27	2018-04-27	NULL	2
4	1590	2018-04-27	2018-04-27	NULL	1
5	1584	2018-04-26	2018-04-26	NULL	2
6	1585	2018-04-26	2018-04-26	NULL	1
7	1582	2018-04-25	2018-04-25	NULL	1
8	1575	2018-04-23	2018-04-23	NULL	2
9	1564	2018-04-21	2018-04-21	NULL	1

Ese... | CSP-2527-14 (17.0 RTM) | EDU-ITS\luca.tarallo (51) | BikeStores | 00:00:00 | Riga: 1, Colonna: 1 | Righe 184

5. Visualizzare il numero di prodotti venduti in un certo negozio in un determinato anno e un certo impiegato

- Stored Procedure :

```
Create procedure sales.usp_GetSalesByStoreYearStaff
    @store_name varchar(100),
    @year       int,
    @last_name  varchar(100)
as
begin
    Select
        s.store_name,
        year(o.order_date) as anno,
        st.first_name + ' ' + st.last_name as staff_name,
        sum(oi.quantity) as total_products_sold
    from sales.orders      o
    join sales.order_items oi on o.order_id    = oi.order_id
    join sales.stores      s  on o.store_id    = s.store_id
    join sales.staffs      st on o.staff_id    = st.staff_id
    where s.store_name   = @store_name
      and year(o.order_date) = @year
      and st.last_name   = @last_name
    group by
        s.store_name,
        year(o.order_date),
        st.first_name,
        st.last_name;
end;
```

- Esecuzione Stored Procedure :

```
exec sales.usp_GetSalesByStoreYearStaff @store_name = 'Santa Cruz Bikes',
                                        @year = 2016,
                                        @last_name  = 'Serrano';
```



6. Selezionare il nome e cognome del cliente, il modello, il prezzo e l'anno di produzione del prodotto, la data dell'ordine di tutti i clienti che hanno acquistato prodotti di una certa categoria individuata tramite il nome della categoria

- Stored Procedure :

```
Create procedure sales.usp_GetCustomersByCategory
    @category_name varchar(100)
as
begin
    Select
        c.first_name,
        c.last_name,
        p.product_name,
        p.list_price,
        p.model_year,
        o.order_date
    from sales.customers c
    join sales.orders o on c.customer_id  = o.customer_id
    join sales.order_items oi on o.order_id     = oi.order_id
    join production.products p on oi.product_id  = p.product_id
    join production.categories cat on p.category_id  =
cat.category_id
    where cat.category_name = @category_name
    order by c.last_name, c.first_name, o.order_date;
end;
```

- Esecuzione Stored Procedure :

```
exec sales.usp_GetCustomersByCategory @category_name = 'Mountain Bikes';
```

Risultati | Messaggi

	first_name	last_name	product_name	list_price	model_year	order_date
1	Ester	Acevedo	Trek Remedy 9.8 27.5 - 2018	4999.99	2018	2018-03-07
2	Elinore	Aguilar	Trek Remedy 29 Carbon Frameset - 2016	1799.99	2016	2016-08-23
3	Elinore	Aguilar	Heller Shagamaw Frame - 2016	1320.99	2016	2016-08-23
4	Elinore	Aguilar	Trek Fuel EX 8 29 - 2016	2899.99	2016	2016-11-13
5	Elinore	Aguilar	Surly Big Dummy Frameset - 2017	999.99	2017	2018-04-10
6	Elinore	Aguilar	Trek Fuel EX 9.8 29 - 2017	4999.99	2017	2018-04-10
7	Janetta	Aguirre	Surly Karate Monkey 27.5+ Frameset - 2017	2499.99	2017	2017-12-31
8	Barry	Albert	Surly ECR 27.5 - 2018	1899.00	2018	2018-03-16
9	Cher	Alston	Trek X-Caliber 7 - 2018	919.99	2018	2018-01-02

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7. Definire un elenco per determinare quanti sono i prodotti in Pending, in Processing, in Rejected e in Completed (es. 182 Pending, ... Processing, ... Rejected, ... Completed). Si consideri a tal proposito l'aggiunta di una tabella Order_Status con i riferimenti agli stati Pending=1, Processing=2, Rejected=3, Completed=4

   - Stored Procedure :

```
Create procedure sales.usp_GetOrderCountByStatus
as
begin
    Select
        Case o.order_status
            when 1 then 'Pending'
            when 2 then 'Processing'
            when 3 then 'Rejected'
            when 4 then 'Completed'
            else 'Unknown'
        end as [Order Status],
        sum(oi.quantity) as [Total Products]
    from sales.orders o
    join sales.order_items oi on o.order_id = oi.order_id
    group by o.order_status;
end;
```

   - Esecuzione Stored Procedure :

```
exec sales.usp_GetOrderCountByStatus;
```

Risultati | Messaggi

	Order Status	Total Products
1	Rejected	180
2	Pending	274
3	Completed	6318
4	Processing	306

Esecu... | CSP-2527-14 (17.0 RTM) | EDU-ITS\luca.tarallo (51) | BikeStores | 00:00:00 | Riga: 1, Colonna: 1 | Righe 4